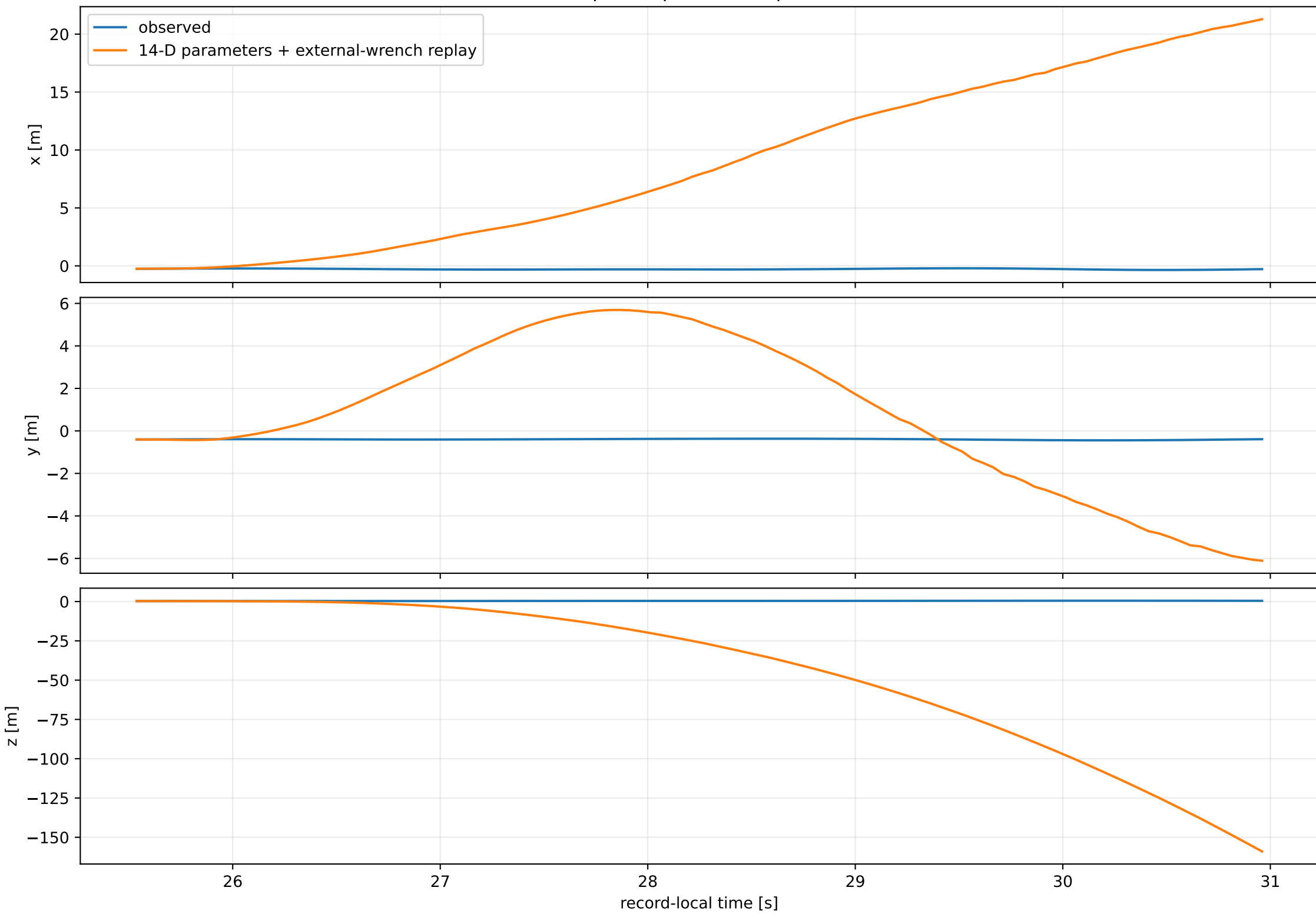
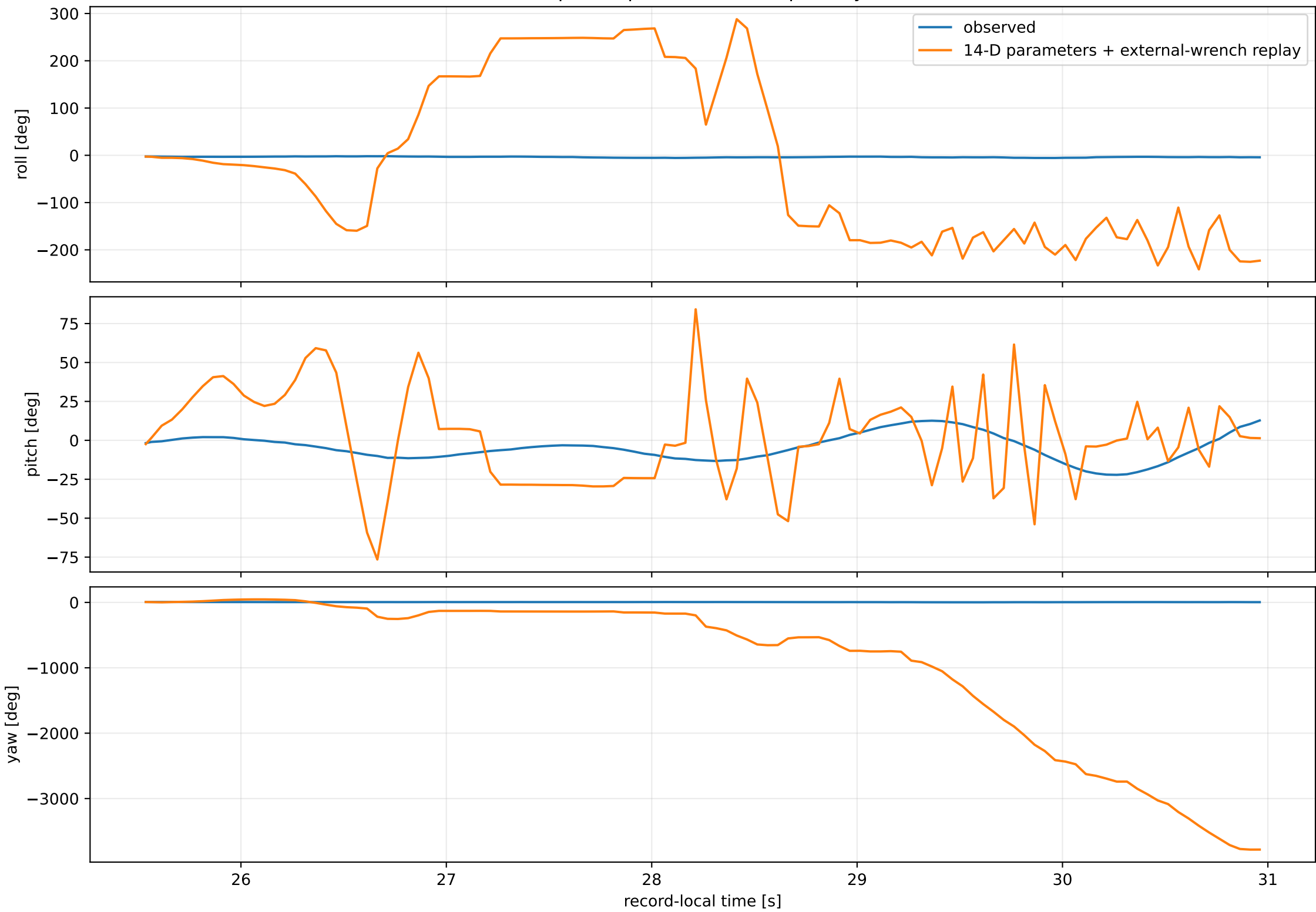


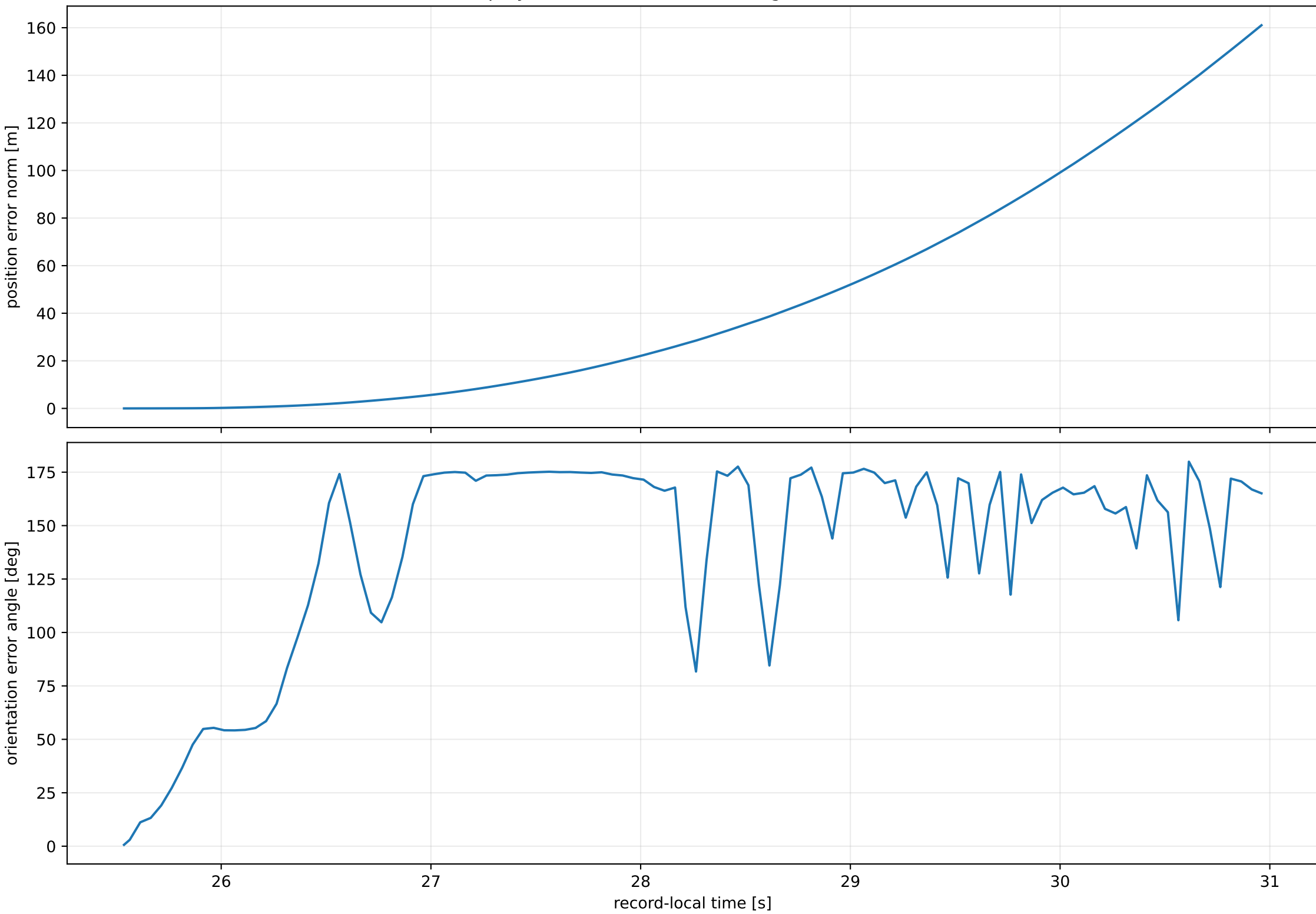
Fine pose reproduction: position



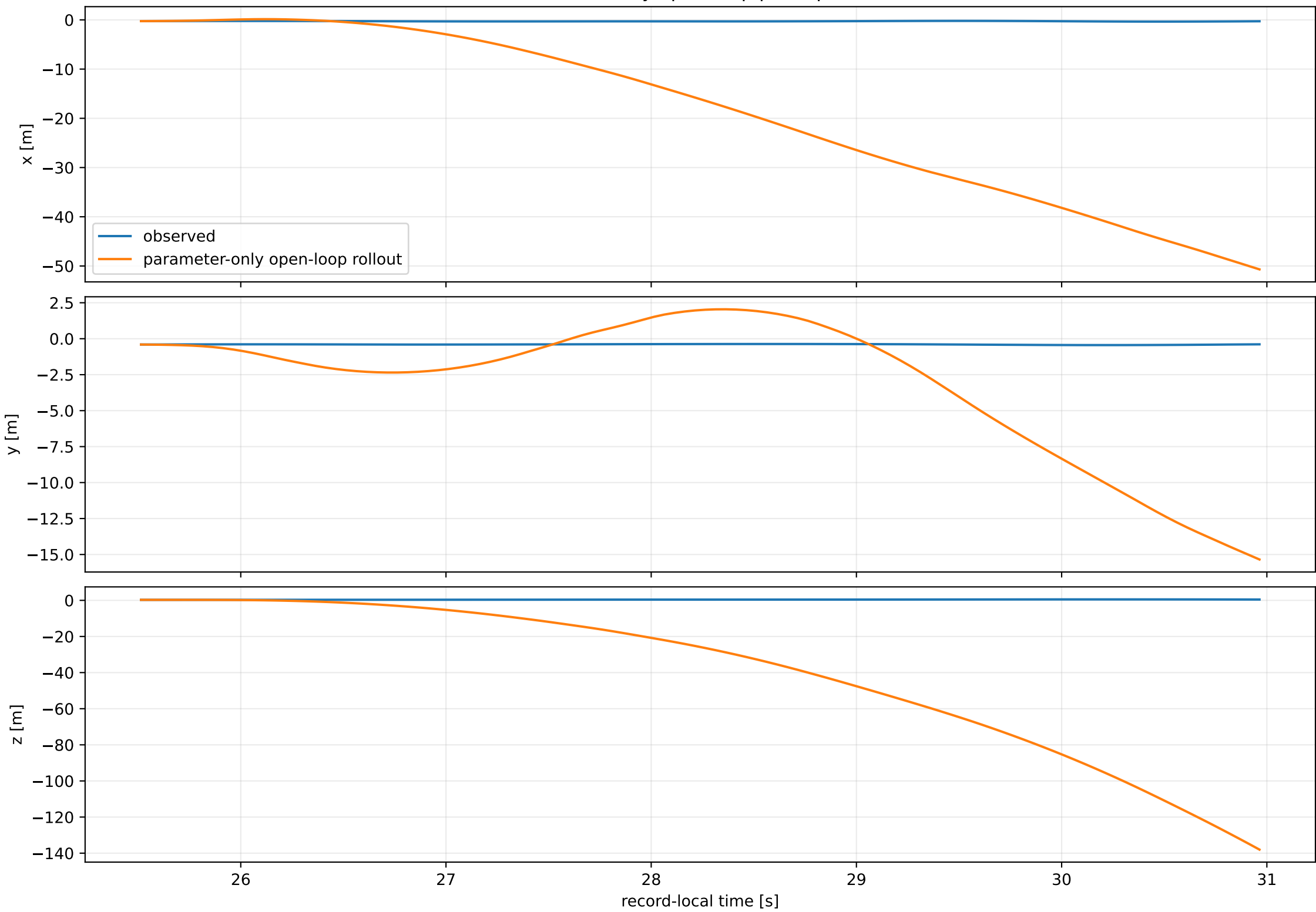
Fine pose reproduction: roll / pitch / yaw



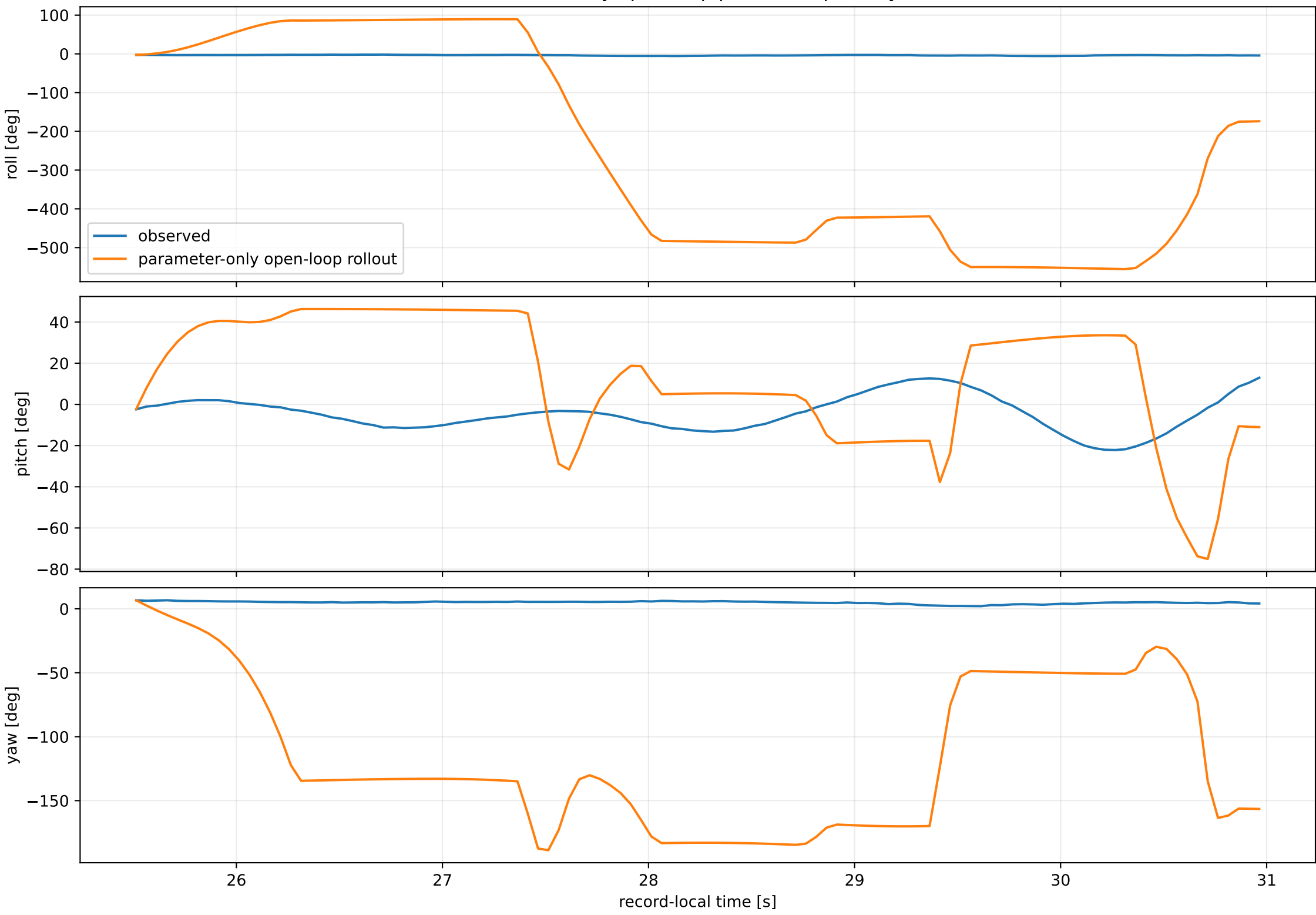
Fine replay error after the 14-D change — lower is better



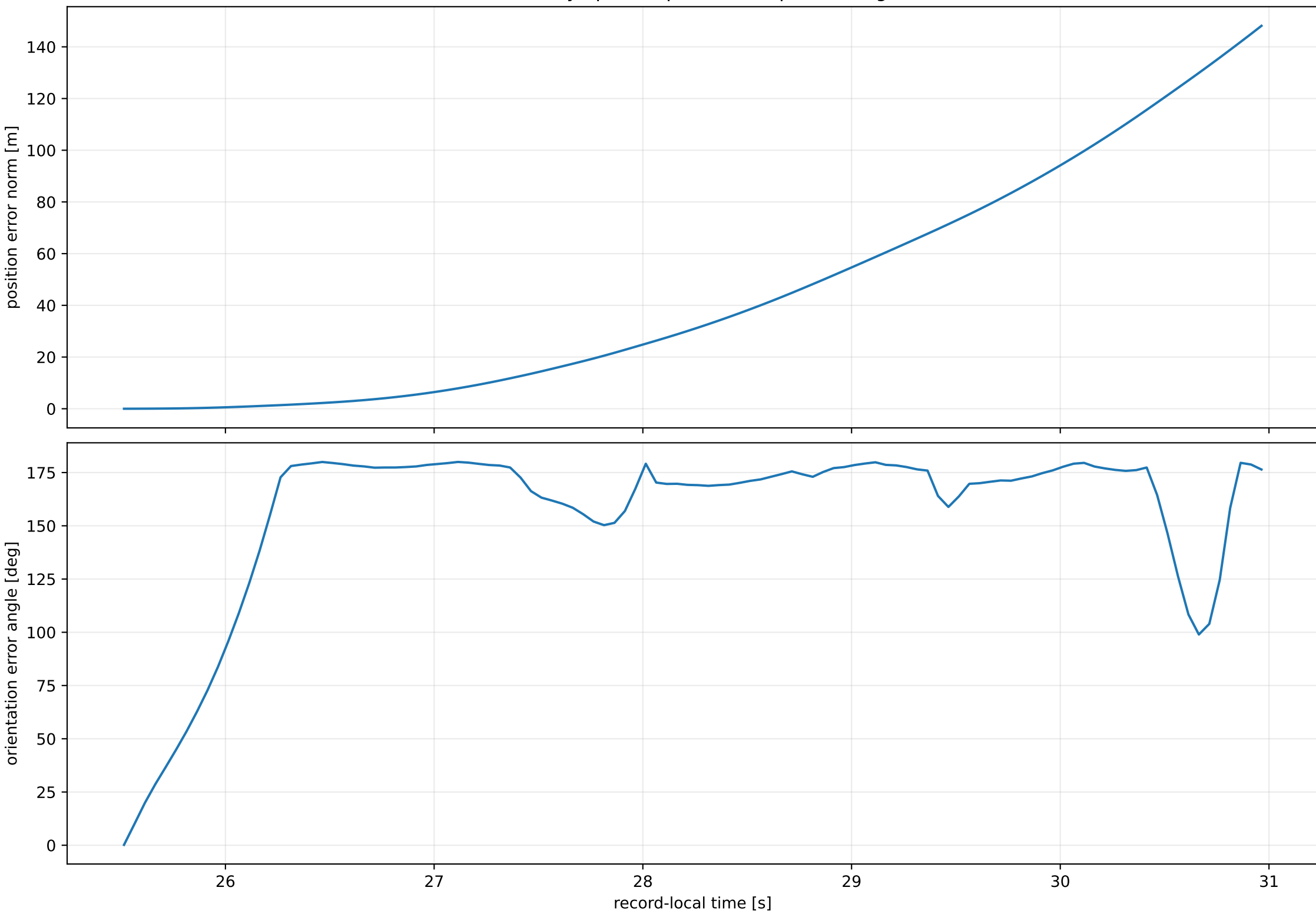
Parameter-only open-loop pose: position



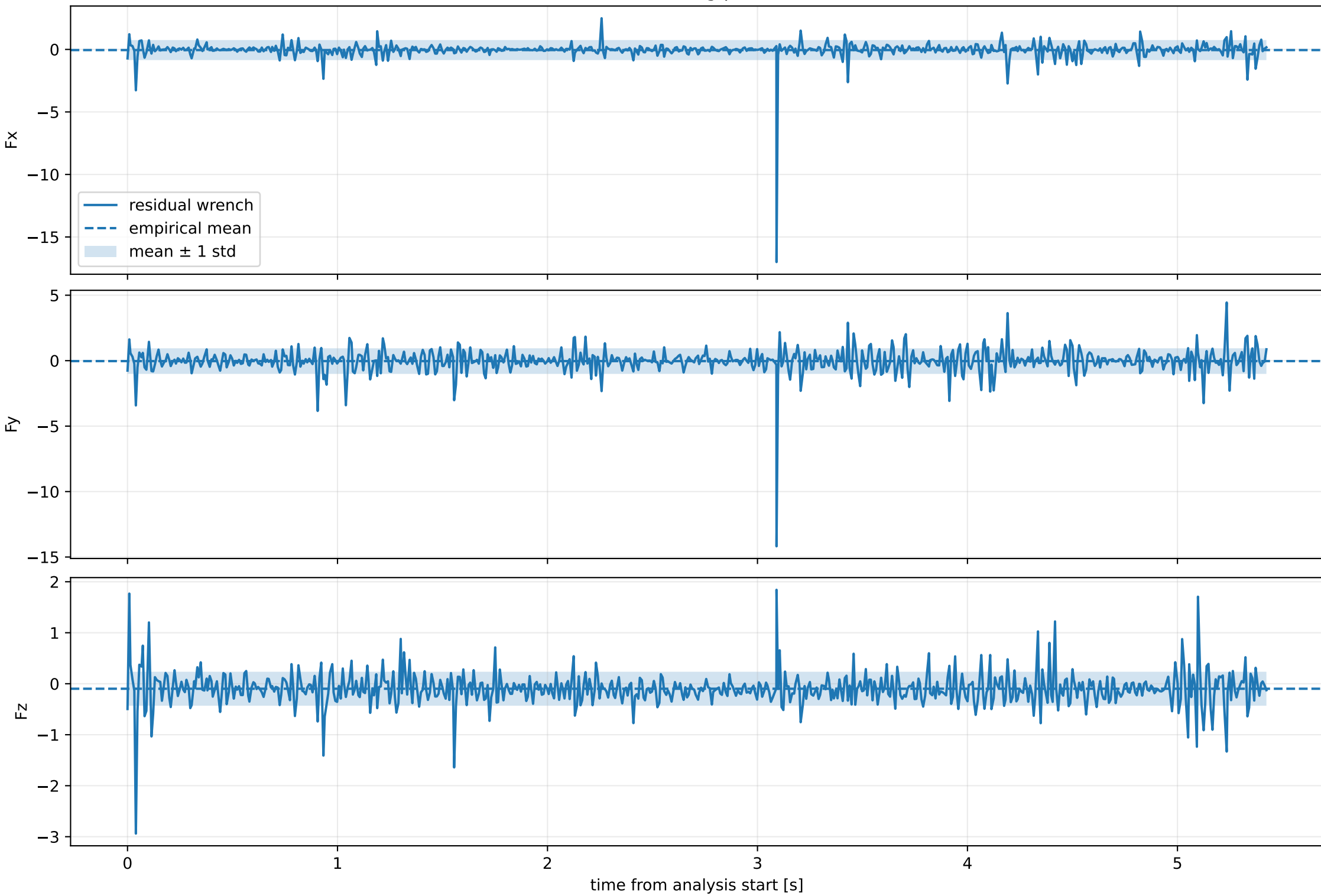
Parameter-only open-loop pose: roll / pitch / yaw



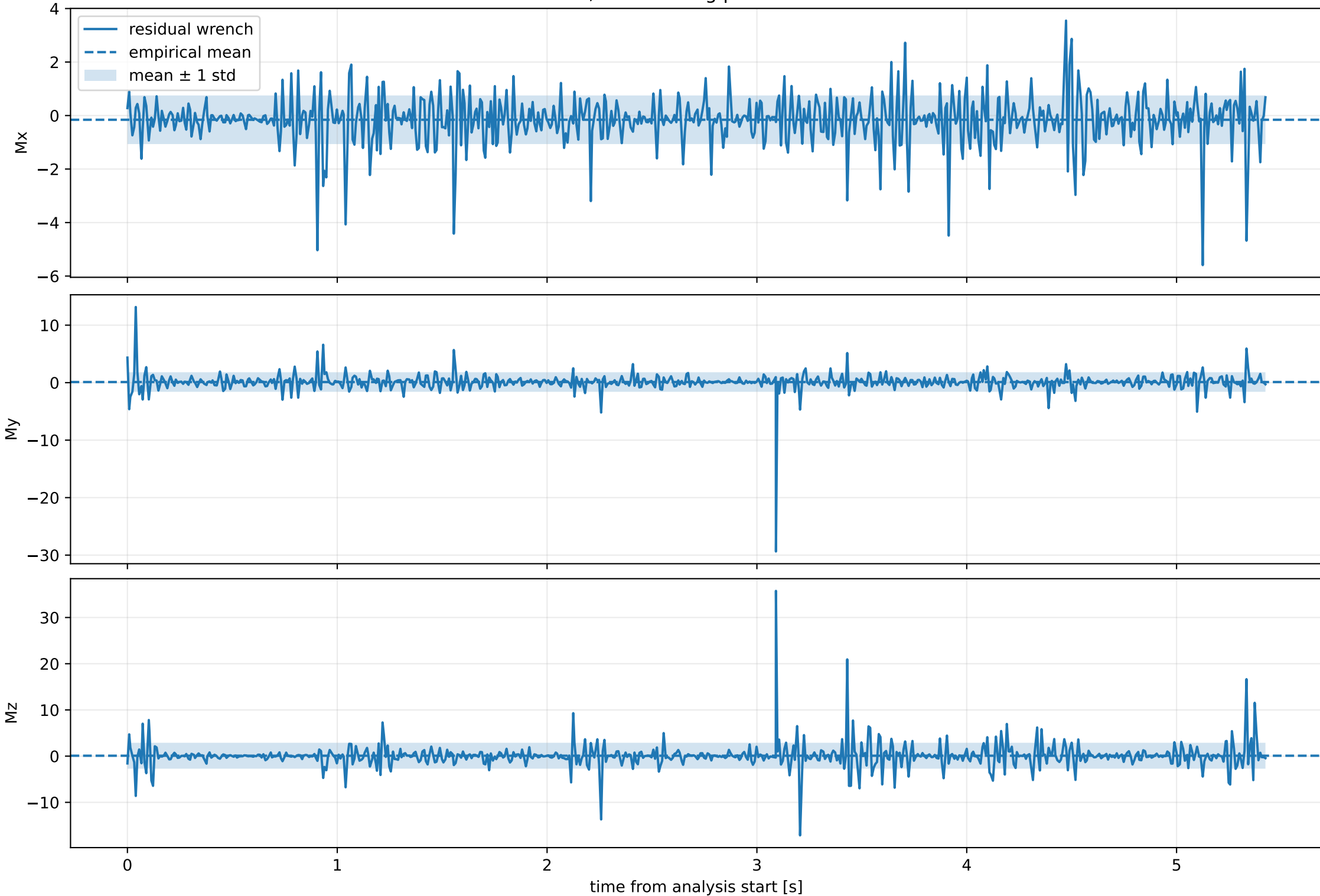
Parameter-only open-loop rollout (separate diagnostic)



Dimensionless residual force: empirical mean  $\pm$  1 std  
Wide band = disturbance varies more, so matching parameter directions receive weaker evidence

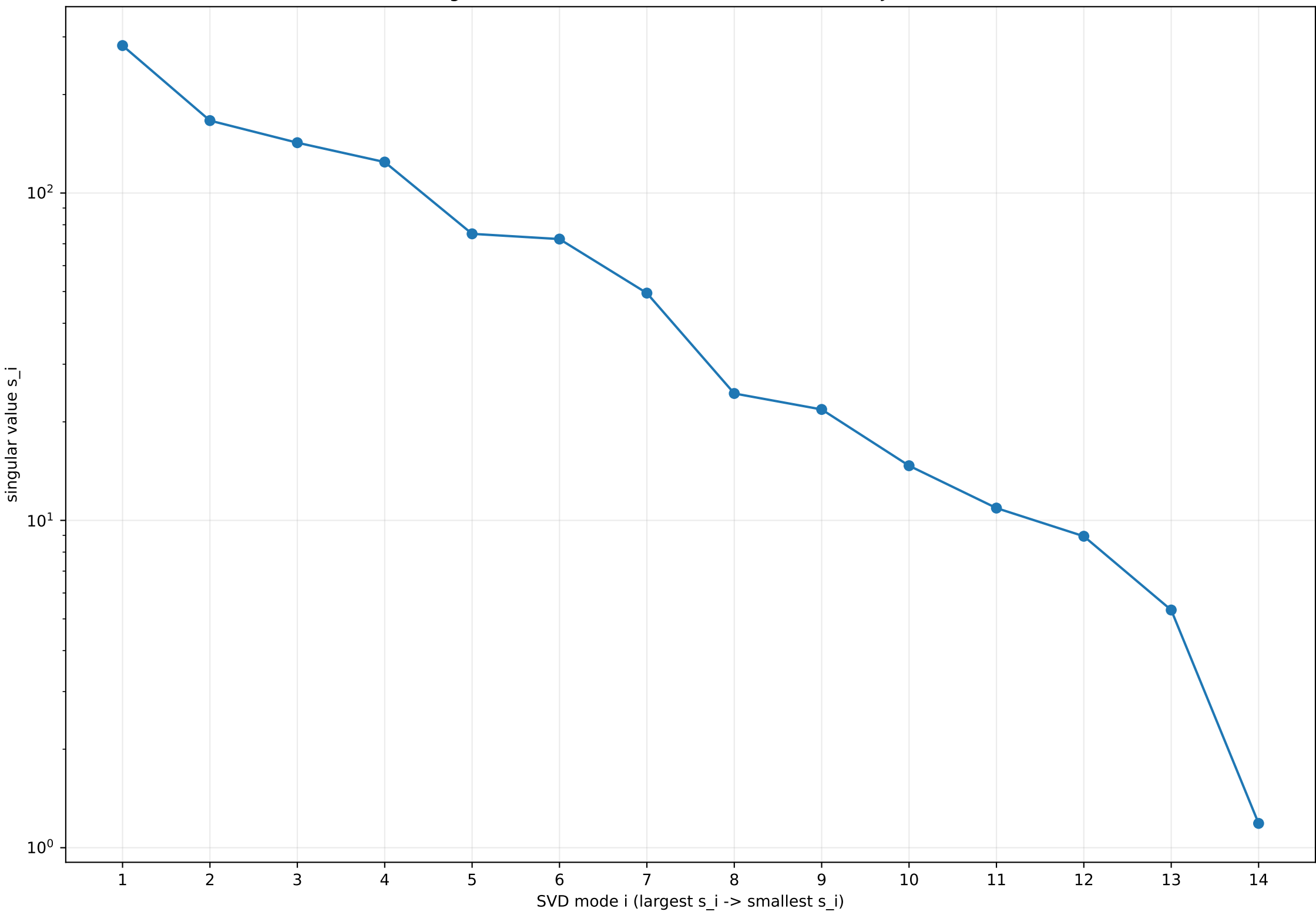


Dimensionless residual torque: empirical mean  $\pm 1$  std  
Wide band = disturbance varies more, so matching parameter directions receive weaker evidence

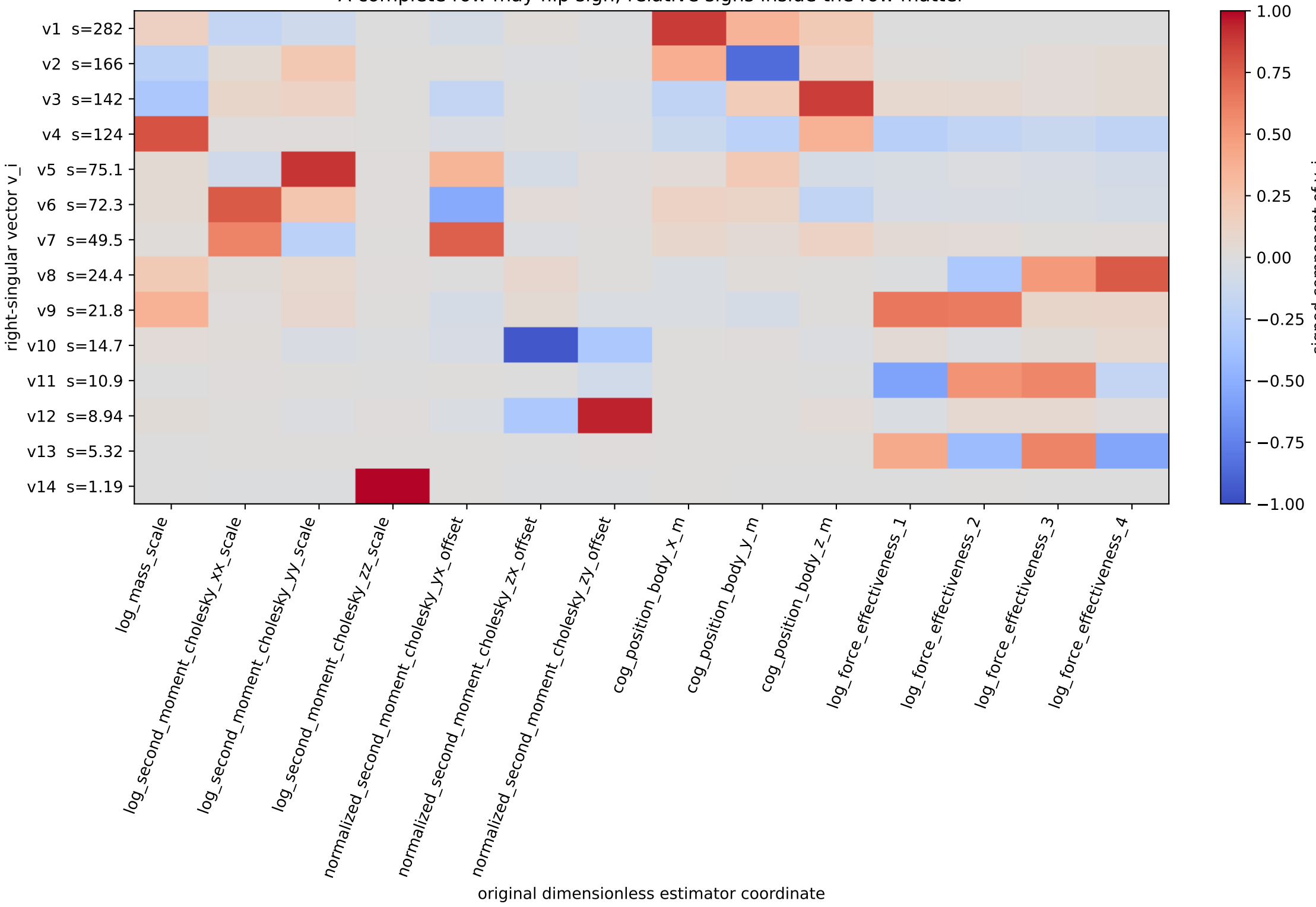




Singular values  $S$  of the whitened dimensionless Jacobian



Signed right-singular vectors  $V^T$   
A complete row may flip sign; relative signs inside the row matter



Decoded physical parameters at the selected estimate

mass [kg]

2.36324308636

inertia J [kg m<sup>2</sup>]

[ 0.065997245 -0.00087107285 -0.00085794595]  
[-0.00087107285 0.065585985 0.00038136934]  
[-0.00085794595 0.00038136934 0.12897466]

principal moments [kg m<sup>2</sup>]

[ 0.064895744, 0.066673362, 0.12898878]

CoG position in body frame [m]

[-0.033813533, -0.037211826, 0.011622842]

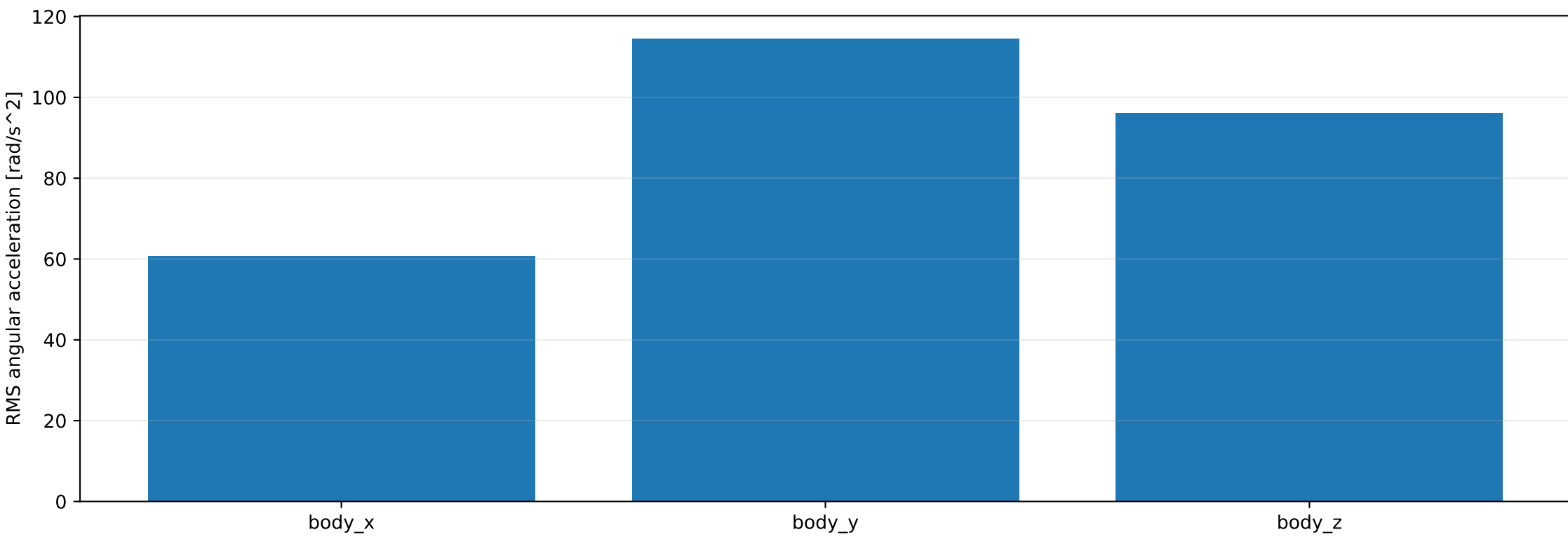
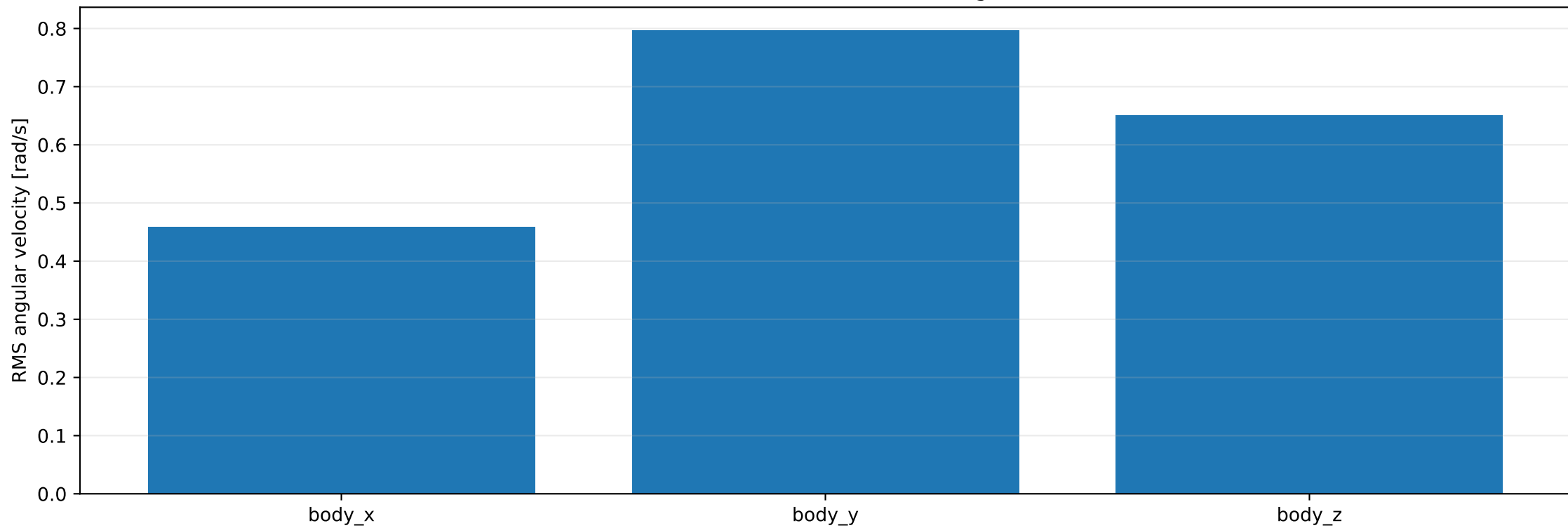
rotor force effectiveness

[ 1.0624831, 0.9704751, 0.85013178, 0.93229841]

selected delay [s]

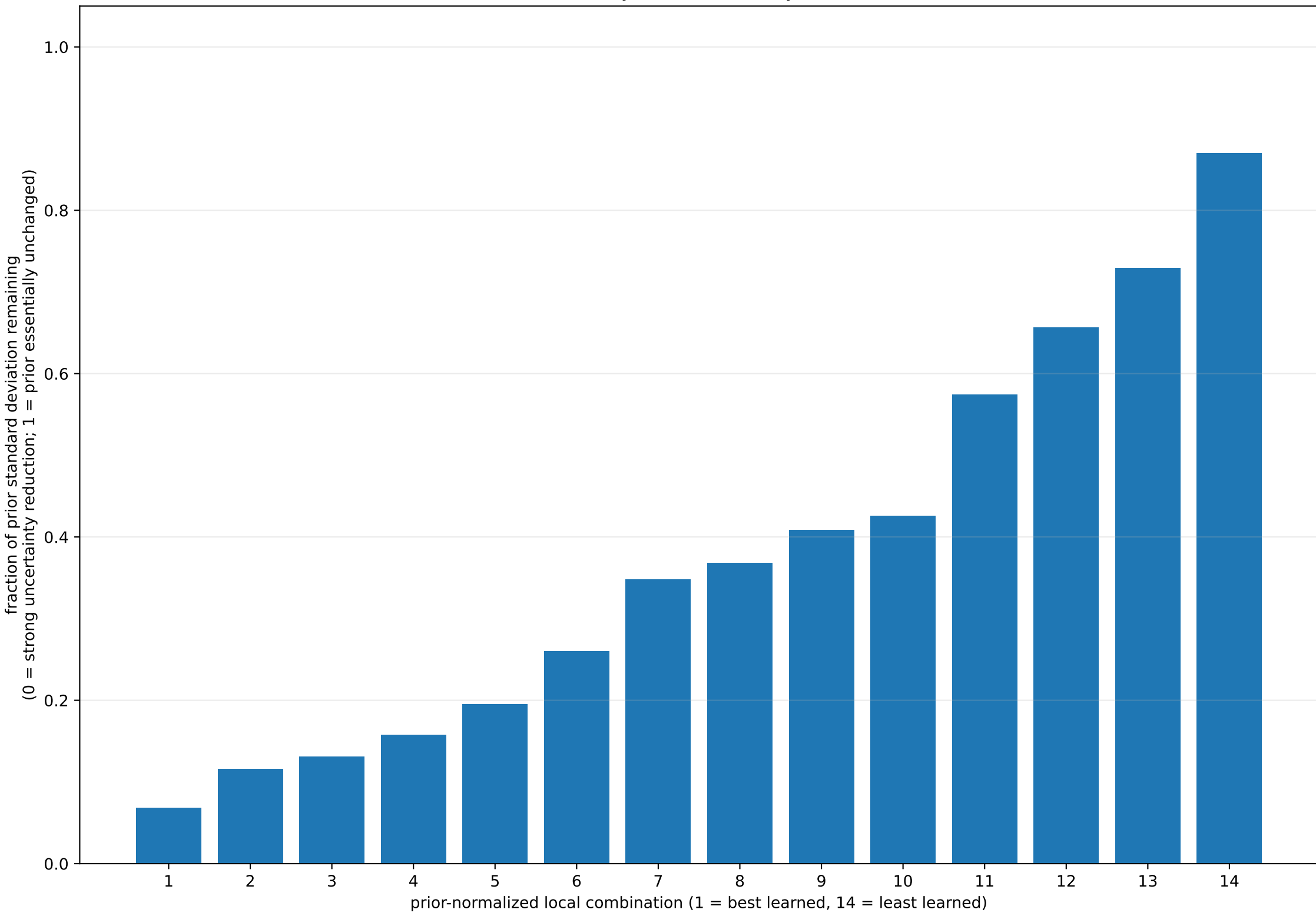
0.004

Axis-wise rotational excitation diagnostic



Low body-z values can support a yaw-excitation hypothesis, but identifiability is decided by the full Jacobian.

How much uncertainty remains in every local combination?



Confidence accumulation as the rosbag becomes longer

